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Case 32-2021: A 14-Year-Old Girl with Swelling of the Jaw and Hypercalcemia

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PRESENTATION OF CASE

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Dr. Austin Be (Surgery): A 14-year-old girl was admitted to this hospital because of swelling of the left jaw.

The patient had been well until 6 weeks before this admission, when she noticed a lump in the left lower jaw; there was no pain, tenderness, or discomfort in the jaw. She was evaluated in a primary care clinic of another hospital and was told that a salivary gland may be obstructed. Treatment with sialagogues, increased oral hydration, warm compresses, and massage of the salivary glands were recommended. During the next 3 weeks, the lump in the left jaw increased in size, and swelling of the left side of the face and neck developed. Three weeks before this admission, she returned to the primary care clinic; treatment with cephalexin was started.

Eighteen days before this admission, the patient was evaluated in the otolaryngology clinic of the other hospital. The temperature was 37.2°C, the blood pressure 132/90 mm Hg, the heart rate 86 beats per minute, and the body-mass index (the weight in kilograms divided by the square of the height in meters) 21.0. There was swelling in the left mandibular and submandibular areas, with no erythema, fluctuance, or palpable salivary stones.

Dr. Hillary R. Kelly: Computed tomography (CT) of the face and neck was performed after the administration of intravenous contrast material. CT of the face (Fig. 1) revealed a 3.7-cm mass in the left mandibular body, with lytic and expansile features and numerous internal wavy septations. Areas of marked thinning and obliteration were noted in the buccal and lingual cortices of the mandible, and extension of the mass among the roots of the first molar, second molar, and premolar was observed. No fluid–fluid or blood–fluid levels were seen within the lesion. There was no lymphadenopathy. The patient was referred to the otolaryngology clinic at this hospital.

Dr. Be: Fifteen days before this admission, the patient reported numbness and tingling in the lower lip and chin on the left side. She had had no fatigue, fever, chills, or redness and no pain in her face, jaw, or neck. On examination, she appeared

well but had obvious facial asymmetry with swelling of the left mandible (Fig. 2). There was a firm, mildly tender mass in the body of the left man-

dible that extended to the mandibular angle. Orthodontic braces were in place and dentition and jaw occlusion were normal. The remainder of the

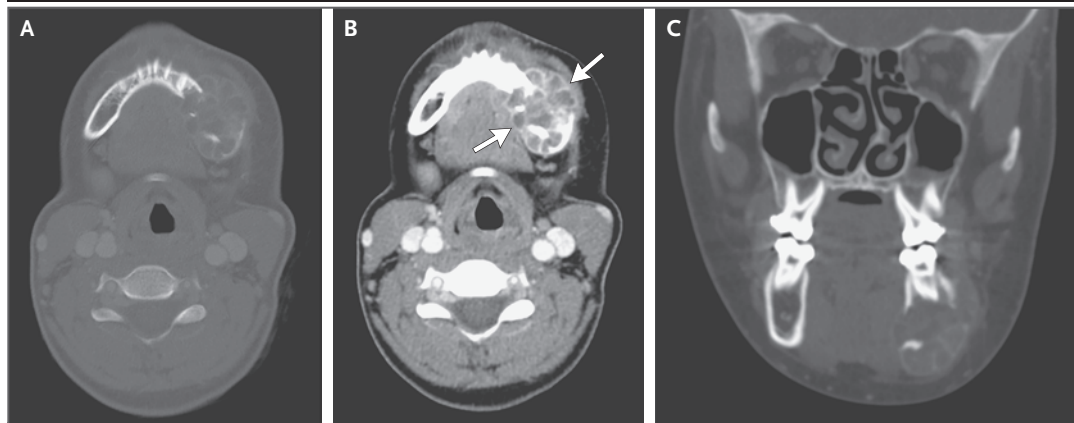


Figure 1. CT Images of the Face.

Axial images obtained after the administration of intravenous contrast material and reconstructed in bone algorithm (Panel A) and in soft-tissue algorithm (Panel B) show an expansile, multiloculated mass centered in the left mandibular body (Panel B, arrows) with internal wavy septations. A coronal image reconstructed in bone algorithm (Panel C) shows defects in the buccal and lingual cortices of the mandible with involvement of the alveolar process and the left mandibular first molar.



Figure 2. Clinical Photographs before Excision of the Lesion.

Frontal and submental views (Panels A and B, respectively) show the expansion of the left mandibular body laterally and inferiorly. Five days later, at the evaluation in the oral and maxillofacial surgical clinic of this hospital, the lesion had enlarged (Panels C and D).

ear, nose, and throat examination was normal. The overlying skin was normal, and no anterior or posterior submandibular or supraclavicular lymphadenopathy was present. The blood levels of sodium, potassium, and chloride were normal, as were the complete blood count and the results of tests of coagulation and renal function. The calcium level was 11.8 mg per deciliter (3.0 mmol per liter; reference range, 8.5 to 10.5 mg per deciliter [2.1 to 2.6 mmol per liter]), and the albumin level was 4.1 g per deciliter (reference range, 3.3 to 5.0). Radiography of the chest was normal.

Dr. Vanía Nosé: An intraoral biopsy of the left mandibular mass was performed. Pathological examination (Fig. 3C) of the biopsy specimen revealed that the lesion was partially surrounded by reactive bone trabeculae, was hemorrhagic, and contained numerous osteoclast-type giant cells around extravasated red cells, findings that were consistent with a central giant-cell lesion or a central giant-cell granuloma.

Dr. Be: Ten days before this admission, the patient was seen in the oral and maxillofacial surgical clinic of this hospital, and admission for the excision of the left mandibular mass was planned.

On admission to this hospital, additional history was obtained. The patient had normal growth and development and had reportedly received all routine childhood vaccinations. She had finished the prescribed course of cephalexin and had taken no other medications; penicillin had caused fever and hives early in childhood. She lived in a suburban area of New England with her mother, stepfather, and two cats; she frequently stayed in another suburban area of New England with her father. The patient performed well in high school and participated in theater. She did not smoke tobacco, use illicit drugs, or drink alcohol. Her mother had hypothyroidism; her father had obstructive sleep apnea. Her maternal grandmother had a history of hypothyroidism, parathyroid adenoma, breast cancer, and colon cancer; her paternal grandfather had died from colon cancer.

The temperature was 36.6°C, the blood pressure 128/85 mm Hg, and the heart rate 111 beats per minute. The physical examination was unchanged. The left mandibular mass was excised. Pathological examination of the surgical specimen revealed abundant giant cells.

On the second hospital day, the calcium level was 12.6 mg per deciliter (3.2 mmol per liter)

and the parathyroid hormone (PTH) level was 396 pg per milliliter (reference range, 10 to 60).

A diagnosis was made.

DIFFERENTIAL DIAGNOSIS

Dr. Thomas O. Carpenter: This 14-year-old girl presented with progressive swelling of the jaw. The salient features of her clinical presentation include swelling of the jaw over the course of 6 weeks; the absence of pain, which implies a recently acquired process; and paresthesias of the lip and chin, which suggest compression of the inferior alveolar nerve. Laboratory test results obtained just before the initial biopsy were notable for an elevation in the blood calcium level, a finding that had not previously been identified. The family history revealed the maternal grandmother's previous diagnoses of parathyroid adenoma, breast cancer, and colon cancer.

This patient's clinical presentation reveals two problems: a mandibular mass and unexplained hypercalcemia. I will develop a differential diagnosis for each of these findings and will consider the ways in which these two conditions may be related.

MANDIBULAR MASS

CT of the face and neck reveals a lytic, expansile lesion with thinning and partial obliteration of cortical bone. Extension of the lesion to the adjacent molars and premolars is described. The presence of abundant giant cells on examination of the biopsy specimen suggests a diagnosis of a giant-cell tumor or granuloma, but the possibility of other diagnoses, such as an aneurysmal bone cyst, a brown tumor, ossifying fibroma, or lytic metastases from cancer, should be considered. Indeed, hypercalcemia may accompany cancer; however, the patient had been completely healthy before the development of jaw swelling, with no other signs of a systemic process, which indicates that the presence of lytic metastases is unlikely. Other processes that should be considered in this case, although they are unlikely, include fibrous dysplasia of bone and multiple myeloma, both of which usually have a radiographic appearance that is considerably different from that observed in this patient and are not characteristically associated with giant cells.

Among the lesions that one might initially consider in this case is a giant-cell tumor of bone.

However, the characteristics commonly associated with such a tumor are inconsistent with this patient's reported clinical history, such that it is typically aggressive and painful, and it occurs more commonly in the appendicular skeleton.¹ Immunohistochemical analysis or genetic testing of the tumor may identify evidence of mutations in *H3F3A*, which occur in more than 95% of patients with such tumors.²

An aneurysmal bone cyst can be manifested by thinning of the bone cortex. However, such a cyst is often characterized by loculated regions and fluid levels, neither of which has been described in this case, and it is not associated with the presence of giant cells.³ Identification of a somatic genetic rearrangement of *USP6*, albeit unlikely, would confirm this diagnosis.⁴

Ossifying fibroma often occurs in the mandible, but its radiographic appearance differs from that described in this patient; it is usually characterized by intact cortexes and the ground-glass appearance of fibrous replacement within the lesion. Giant cells are not typical of an ossifying fibroma, although it has been reported that the tumor may arise from mesenchymal cells of periodontal ligaments, which could be related to the dental findings in this case.⁵

Finally, consideration of a brown tumor in this patient is supported by the lytic appearance and disruption of cortical bone.⁶ Furthermore, giant cells can be present in brown tumors. A brown tumor is a manifestation of chronic exposure to moderately-to-severely elevated levels of PTH, which may be the cause of this patient's hypercalcemia. Several of these tumors have similar histologic features and may at times be difficult to distinguish from one another.

HYPERCALCEMIA

The second feature of this patient's presentation is hypercalcemia. The first step in the evaluation of an abnormal calcium level is to confirm that the initial level of blood calcium is accurate and to ensure that the ionized or corrected calcium level (i.e., the calcium level corrected in the context of a low blood albumin level) is normal. The next step in evaluating hypercalcemia is to establish its cause. It is useful to consider mechanisms that would explain the hypercalcemia. Is it due to excessive bone resorption or lytic metastases (relevant in this case with respect to consideration of the jaw mass)? Is there excessive intestinal

absorption of calcium? Is it a combination of these mechanisms?

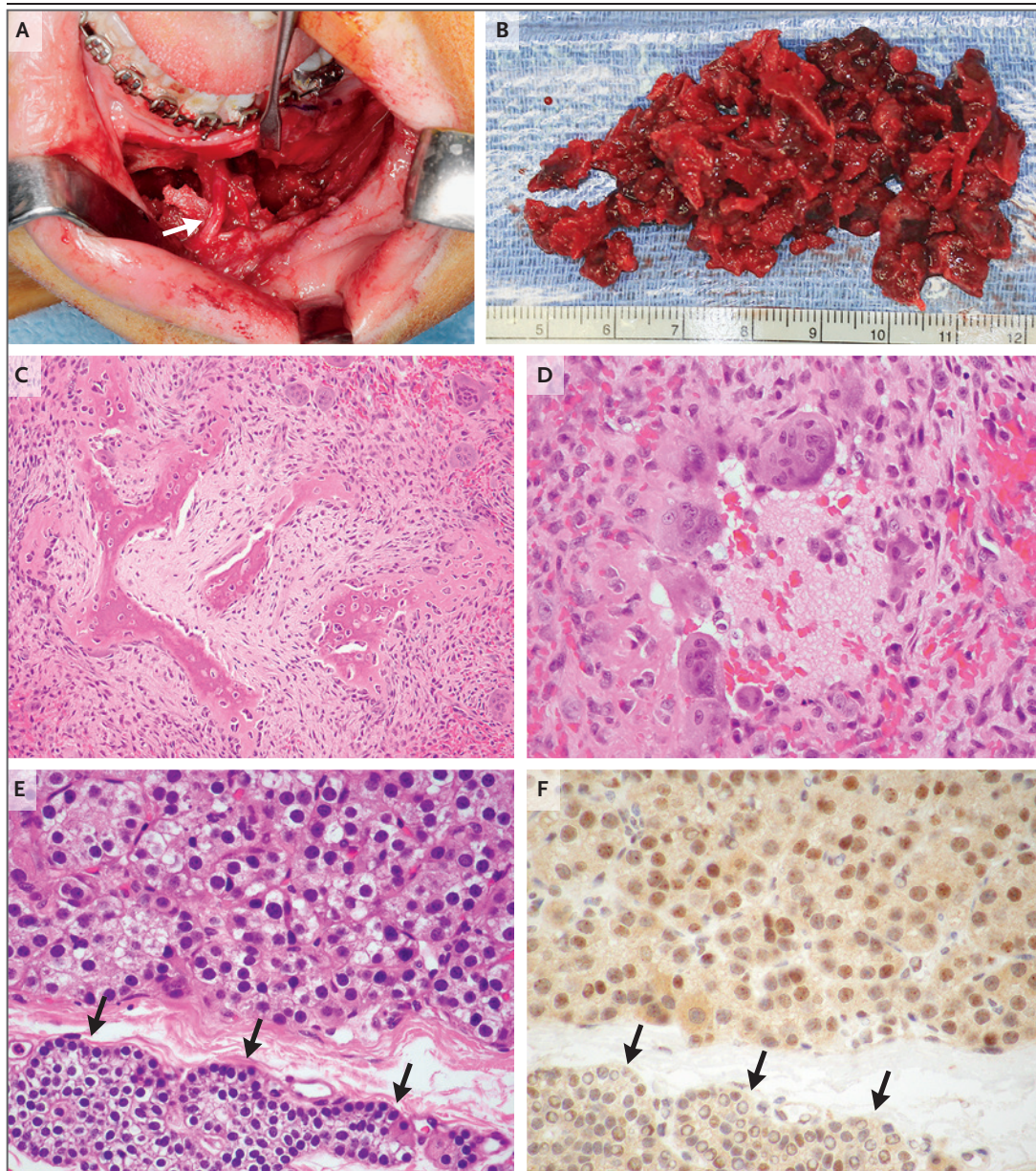
HYPERCALCEMIA AND PTH LEVEL

Is this patient's elevated calcium level mediated by PTH? Non-PTH-mediated bone resorption may occur in patients with hypercalcemia that is associated with cancer or one of several inflammatory diseases (possibly due to cytokine-induced bone resorption), usually causing low circulating PTH levels. Hypervitaminosis D and defects in vitamin D metabolism (caused by mutations in *CYP24A1*, the gene encoding the vitamin D 24-hydroxylase)^{7,8} are typically manifested by hypercalcemia and suppressed PTH levels. Defects in renal phosphate transport can have secondary effects of increasing production of the active vitamin D metabolite 1,25-dihydroxyvitamin D, thereby resulting in hypercalcemia from both increased intestinal calcium absorption and increased bone resorption, with low circulating PTH levels.^{9,10}

When this patient was admitted to the hospital, the blood calcium level was further elevated and the circulating PTH level was markedly elevated, findings that confirm the diagnosis of primary hyperparathyroidism. Although this condition is probably caused by a solitary parathyroid adenoma, multiglandular parathyroid hyperplasia or multiple adenomas may be present. A concomitant elevation in the blood alkaline phosphatase level is evidence of the sustained effect of PTH on osteoblasts.

FAMILIAL HYPERPARATHYROIDISM SYNDROMES

Given this patient's family history of parathyroid adenoma in her grandmother, familial hyperparathyroidism syndromes — particularly multiple endocrine neoplasia type 1, type 2, and type 4¹¹⁻¹³ and hyperparathyroidism–jaw tumor syndrome¹⁴ — should be considered. In persons older than this patient, multiple endocrine neoplasia is usually manifested by hyperparathyroidism and is typically associated with multiglandular disease. Hyperparathyroidism–jaw tumor syndrome, which is caused by germline mutations in *CDC73* (also known as *HRPT2*), is characterized by solitary or multiple, often asynchronous, parathyroid adenomas. These tumors may have aggressive pathological features such as fibrous bands or trabeculation, while still being classified as adenomas.¹⁵ Immunohistochemical staining of a biopsy spec-



imen would indicate the absence of parafibromin, a protein encoded by *CDC73*. Accompanying jaw tumors, which are typically described as ossifying fibromas, may or may not be present; in the absence of jaw tumors, the condition can be described as familial isolated hyperparathyroidism. Nevertheless, uterine tumors or endometrial hyperplasia, as well as renal cysts, may also occur. Because there is an increased incidence of parathyroid carcinoma among persons with *CDC73* mutations, it is important for affected patients to undergo active lifelong surveillance.¹⁵ Indeed,

patients with *CDC73*-related hyperparathyroidism may also present with brown tumors.¹⁶ This new focus of the current case emphasizes the importance of establishing a genetic diagnosis to determine how best to manage the accompanying parathyroid disease.

To distinguish between a solitary parathyroid adenoma and multiglandular parathyroid hyperplasia, imaging of the parathyroid glands is warranted.¹⁷ Such imaging may also inform the surgeon as to the most appropriate approach for parathyroidectomy. Multiple imaging methods are

Figure 3 (facing page). Surgical Approach and Biopsy and Surgical Specimens.

Enucleation of the mandibular lesion was performed through an intraoral incision in the buccal vestibule lateral to the teeth (Panel A). The lesion was easily separated from the thin, expanded cortical margins. The inferior alveolar nerve (arrow) was preserved. The high vascularity of the tumor imparts a red-brown appearance to the lesion that is typical of brown tumors and other giant-cell-rich lesions (Panel B). Hematoxylin and eosin staining of the initial maxillary tumor-biopsy specimen before excision (Panel C) shows a nodule of the brown tumor that is partially surrounded by trabeculae of reactive bone. The solid tumor is hemorrhagic and contains osteoclasts. Hematoxylin and eosin staining of the maxillary tumor specimen after excision (Panel D) shows that the tumor is hypercellular and is composed of spindle cells admixed with numerous osteoclasts that tend to cluster around areas of extravasated red cells. Approximately 3 weeks after excision of the tumor, parathyroidectomy was performed. Hematoxylin and eosin staining of the left upper parathyroid gland specimen (Panel E) shows an enlarged parathyroid gland that measures 2.4 cm in greatest dimension. The gland weighed 1500 mg. The specimen was hypercellular, with a diffuse growth pattern, and was composed of chief cells with a rim of normocellular parathyroid tissue (Panel E, arrows). Immunohistochemical staining for parafibromin and *CDC73* (Panel F) shows preservation of parafibromin and *CDC73* by the adenomatous cells and by the cells within the rim of normocellular parathyroid tissue (arrows).

available, including radionuclide scanning (with the use of technetium-99m–labeled sestamibi), dynamic CT, magnetic resonance imaging, and ultrasonography. I would opt for ultrasonography in this young patient in order to avoid exposure to the radiation that is inherent with CT and radionuclide scans.

How does the finding of hyperparathyroidism influence the diagnosis of the mandibular mass? The possibility of the diagnosis of an atypical ossifying fibroma is increased, given its occurrence in the context of hyperparathyroidism due to *CDC73* mutations. Nevertheless, brown tumors are a complication of hyperparathyroidism, and the reported biopsy and radiographic findings are more consistent with this type of lesion.

**DR. THOMAS O. CARPENTER'S
DIAGNOSIS**

Brown tumor associated with primary hyperparathyroidism (probably familial).

ADDITIONAL DIAGNOSTIC TESTS

Dr. Jordan S. Sherwood: Additional laboratory testing performed after the excision of the mandibular mass was notable for a 25-hydroxyvitamin D level of less than 13 ng per milliliter (32 nmol per liter; reference range, 33 to 96 ng per milliliter [82 to 240 nmol per liter]) and a 1,25-dihydroxyvitamin D level of 154 pg per milliliter (370 pmol per liter; reference range, 24 to 86 pg per milliliter [58 to 206 pmol per liter]). The phosphorus level was 2.6 mg per deciliter (0.8 mmol per liter; reference range, 3.0 to 4.5 mg per deciliter [1.0 to 1.5 mmol per liter]). A random urine sample revealed hypercalciuria; the ratio of calcium to creatinine was elevated at 0.8 (reference value, <0.2). These laboratory findings were consistent with PTH-dependent hypercalcemia. Review of the previously obtained imaging study, as well as additional imaging, was performed for disease localization of the primary hyperparathyroidism.

Dr. Kelly: Further review of the CT images of the neck (Fig. 4A and 4B), which had been obtained before the biopsy of the mandibular mass, after the administration of intravenous contrast material, revealed a 2-cm nodule in the left tracheoesophageal groove, immediately posterior to the mid-to-upper pole of the left lobe of the thyroid gland, consistent with a parathyroid adenoma. This parathyroid adenoma was hypoattenuating relative to the adjacent thyroid parenchyma when viewed in a narrow soft-tissue window. An ultrasound image of the neck (Fig. 4C) showed the parathyroid adenoma as a hypoechoic nodule separate from and posterior to the left lobe of the thyroid gland.

**DISCUSSION OF MEDICAL
MANAGEMENT**

Dr. Sherwood: In the context of primary hyperparathyroidism, the lesion in the jaw was thought to be a brown tumor rather than the initially suspected giant-cell tumor. Brown tumors are non-neoplastic giant-cell focal lesions of bone that are associated with hyperparathyroidism and have histologic features similar to those of giant-cell tumors.¹⁸ Brown tumors typically regress after treatment of the underlying hyperparathyroidism with parathyroidectomy, but local surgery is sometimes needed, depending on the location

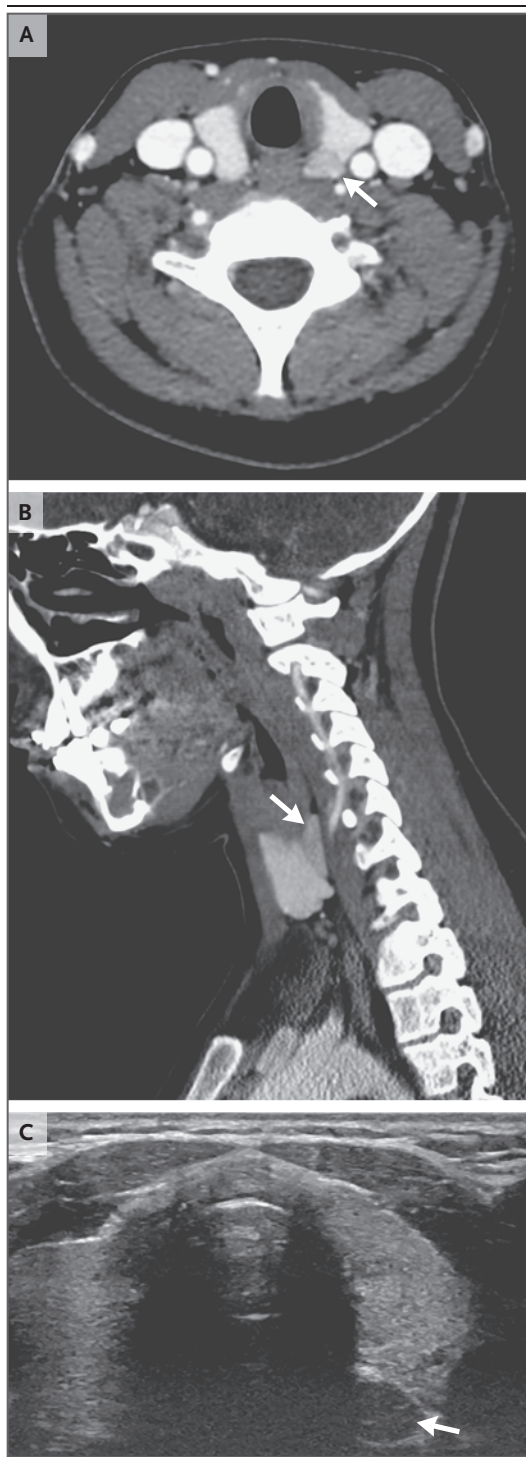


Figure 4. CT and Ultrasound Images of the Neck.

Axial and coronal CT images of the neck (Panels A and B, respectively), obtained after the administration of intravenous contrast material (and at the same time as the CT images of the face), show a nodule (arrows) in the left tracheoesophageal groove, immediately posterior to the mid-to-upper pole of the left lobe of the thyroid gland. This nodule was hypoattenuating relative to the adjacent thyroid parenchyma. A transverse ultrasonographic image of the neck (Panel C) confirms that this hypoechoic lesion (arrow) in the left tracheoesophageal groove is separate from the thyroid gland (and does not arise within the thyroid parenchyma), a feature that is consistent with a parathyroid adenoma.

monitoring may be appropriate in the adult population, depending on the clinical scenario.²⁰ Given this patient's age and overt manifestation of hyperparathyroidism (i.e., the presence of a brown tumor), a parathyroidectomy was scheduled.

While the patient awaited definitive surgical management, her hypercalcemia was treated with hydration. Given the elevated 1,25-dihydroxyvitamin D level, which could result in increased calcium absorption from the gut, a low-calcium diet was recommended.²¹ Vitamin D repletion therapy was initiated preoperatively, since post-surgical hypocalcemia is more common among patients with vitamin D deficiency after parathyroidectomy than among those without vitamin D deficiency.²²

Bone and renal manifestations associated with primary hyperparathyroidism are common and are related to the sustained action of elevated PTH levels. Additional studies were obtained to assess for other consequences of hyperparathyroidism. Dual-energy x-ray absorptiometry (DXA) revealed substantially decreased bone mineral density. The patient had no history of fragility fractures or clinical findings that were suggestive of vertebral fractures. In addition to the urine studies that showed hypercalciuria, renal ultrasonography revealed a 5-mm, echogenic focus in the left lower pole of the left kidney that was consistent with a nonobstructing stone. There were no signs of nephrocalcinosis. The patient was otherwise asymptomatic, without renal colic or polyuria, and the renal function was normal. After definitive surgical management of primary hyperparathyroidism, bone mineral density typically improves and nephrolithiasis usually begins to abate²³; these conditions were followed clini-

and size of the tumor.¹⁹ In the pediatric population, definitive surgical management for the treatment of hyperparathyroidism is usually recommended; in contrast, medical management and

cally in this patient. Given the possibility that this patient had a genetic mutation as the underlying cause of her syndrome, she was referred for genetic counseling and testing.

DISCUSSION OF SURGICAL MANAGEMENT

Dr. Zachary S. Peacock: On the basis of the large size of this patient's tumor and the extent of associated cortical destruction, the lesion had been classified as aggressive. The patient had been offered intraosseous enucleation of the mandibular mass with preservation of the teeth and inferior alveolar nerve, along with adjuvant medical treatment. However, shortly after her initial visit with me, we were notified that the lesion had increased in size, and the patient had begun having increased discomfort and social isolation. Enucleation of the lesion was scheduled for the next day.

Enucleation of the lesion was performed by means of an intraoral incision through the mucosa lateral to the teeth (Fig. 3A and 3B). Postoperatively, the patient was placed on a blenderized diet, and she received zoledronic acid to expedite restoration of bone in the mandibular defect.

PATHOLOGICAL DISCUSSION

Dr. Nosé: Central giant-cell lesions are uncommon and may arise on jaw bones or on small tubular bones of the hands and feet. Such lesions are composed of spindled fibroblasts and collagenous stroma with large areas of hemorrhage and numerous multinucleated osteoclast-type giant cells. The giant cells tend to occur in small clusters around areas of hemorrhage, and each cell usually contains up to 12 nuclei. The two most important diagnoses to consider in this case are brown tumor of hyperparathyroidism and giant-cell tumor of bone.²⁴ The central giant-cell granuloma closely resembles a brown tumor of hyperparathyroidism.²⁵ The giant-cell tumor of bone contains numerous giant cells that have more than 12 nuclei each, and the cells are distributed evenly throughout the tumor.

After the initial biopsy, the left mandibular mass was excised, and the morphologic findings (Fig. 3D) were similar to those of the previous biopsy. The diagnoses of central giant-cell lesion

or central giant-cell granuloma were considered until further clinical information was received. Given this patient's clinical history of hyperparathyroidism, the tumor was ultimately diagnosed as a brown tumor associated with hyperparathyroidism. Central giant-cell lesions, central giant-cell granulomas, and brown tumors associated with hyperparathyroidism all have similar histopathological features.^{24,26}

Approximately 3 weeks after the mandibular tumor excision, the patient underwent left and right upper parathyroidectomy (Fig. 3E and 3F). The left upper parathyroid gland was enlarged, measuring 2.4 cm in greatest dimension and weighing 1500 mg (reference range, 30 to 50). The specimen was hypercellular and had a rim of normocellular parathyroid tissue. The lesion had a diffuse growth pattern and was composed of chief cells. On immunohistochemical staining, tumor cells were positive for chromogranin, PTH, and parafibromin, the gene product of *CDC73*²⁷; the presence of parafibromin ruled out hyperparathyroidism–jaw tumor syndrome or another *CDC73*-dependent disease.²⁸ Moreover, the tumor present in this syndrome is usually ossifying fibroma of the jaw, which is characterized by fibroblast-rich stroma, irregular spicules of woven bone, and osteoblastic rimming that lacks giant cells.

ANATOMICAL DIAGNOSIS

Brown tumor associated with hyperparathyroidism and parathyroid adenoma.

GENETIC TESTING AND FOLLOW-UP

Dr. Sherwood: Genetic testing for mutations in *CDC73*, *RET*, and *CDKN1B* and for multiple endocrine neoplasia did not identify any pathogenic variants. The parathyroidectomy was uncomplicated, and the PTH level normalized immediately. The patient did not have postoperative hypocalcemia and was discharged home without complications. The levels of PTH and calcium have since been serially monitored over time, and both levels have remained within normal limits. Given the severe disruption of the cortical bone of the jaw, she received intravenous bisphosphonate therapy with zoledronic acid on an outpatient basis after the parathyroidectomy. A repeat DXA scan showed marked improvement in bone

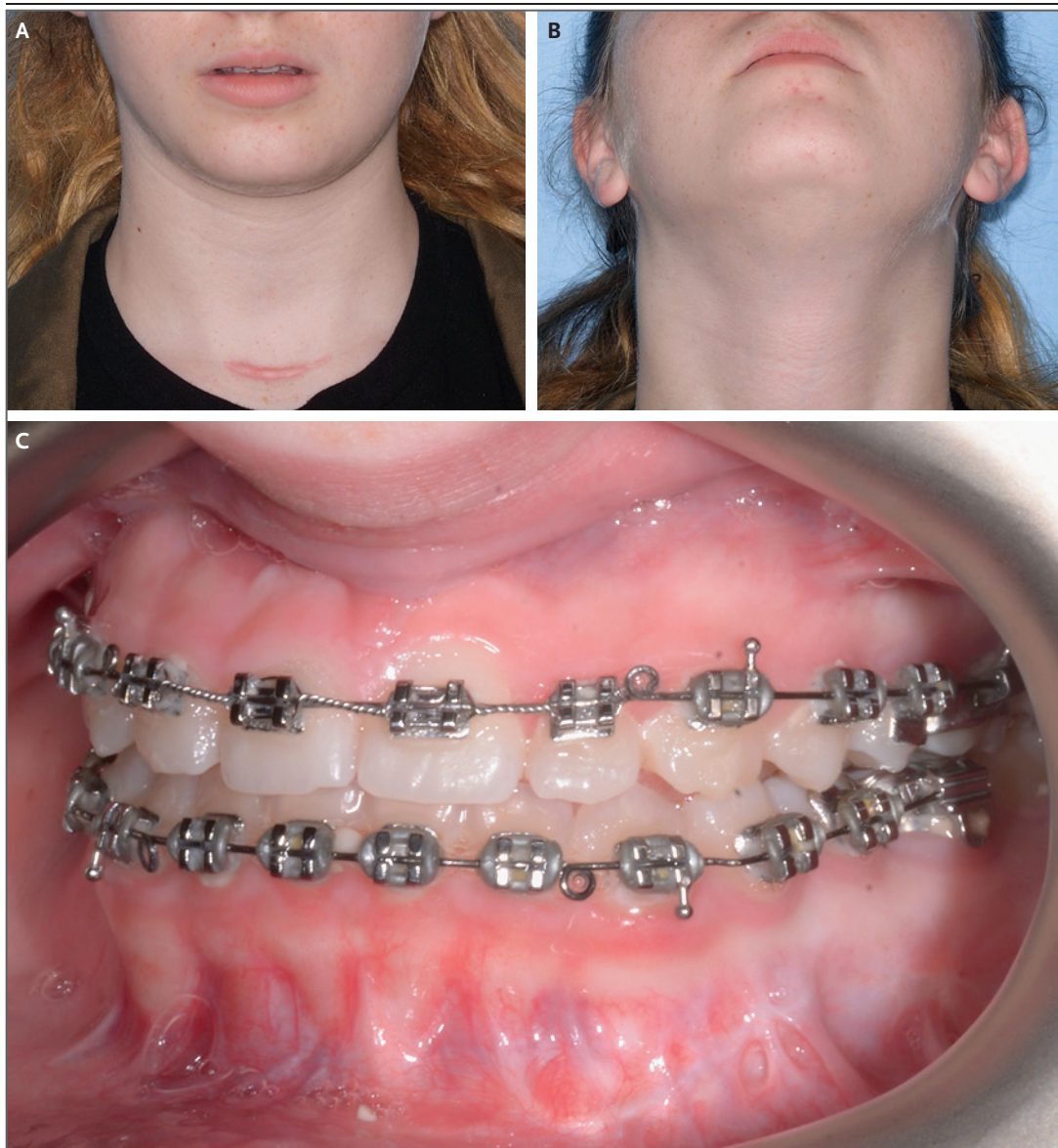


Figure 5. Clinical Photographs 10 Months after Excision of the Lesion.

At a follow-up visit 10 months after the mandibular enucleation, the mandibular defect has ossified, and the native contour has been restored (Panels A and B). The teeth remain in good health, and sensation in the lower lip and chin has returned, which reflects preservation of the inferior alveolar nerve (Panel C).

mineral density. At a follow-up visit 10 months after the surgical procedures, the patient was doing well, and the contour of her mandible was fully restored (Fig. 5). Her teeth were maintained, and the sensation in the lower lip and chin returned.

FINAL DIAGNOSIS

Primary hyperparathyroidism and brown tumor.

This case was presented at Pediatric Grand Rounds.
Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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